java script ki history btao

JavaScript की History (इतिहास)

JavaScript एक बहुत ही लोकप्रिय programming language है, जिसका इस्तेमाल मुख्य रूप से Web Development में किया जाता है। इसकी शुरुआत 1995 में हुई थी।

1. JavaScript की शुरुआत कब और किसने की?

- **Creator:**

Brendan Eich

- **Year:** 1995
- **Company:** Netscape
- **पहला नाम:** Mocha
- **दूसरा नाम:** LiveScript
- **Final name:** JavaScript

1995 में Netscape कंपनी को अपने browser में interactive features जोड़ने के लिए एक scripting language की जरूरत थी।

इसलिए Brendan Eich ने केवल लगभग **10 दिनों में JavaScript का पहला version बनाया था।**

2. JavaScript के नाम का इतिहास

1. **Mocha (May 1995)** JavaScript का सबसे पहला नाम Mocha था।
2. **LiveScript (September 1995)** बाद में इसका नाम LiveScript रखा गया।
3. **JavaScript (December 1995)** फिर इसका नाम JavaScript कर दिया गया।
4. **ECMAScript (1997)** JavaScript के लिए एक official standard बनाया गया, जिसे ECMAScript कहा जाता है।

Important: JavaScript और Java दोनों अलग-अलग programming languages हैं। इनके नाम मिलते-जुलते हैं, लेकिन इनका संबंध ऐसा नहीं है कि JavaScript, Java का छोटा version हो।

3. JavaScript की Important Timeline

Year	Event
1995	JavaScript का जन्म
1996	Microsoft ने JScript बनाया
1997	ECMAScript 1 standard आया
1999	ES3 release हुआ

Year	Event
2009	ES5 release हुआ
2015	ES6 (ES2015) release हुआ
2016	ES2016 release हुआ
2017	ES2017 release हुआ
2020	ES2020 release हुआ
2023	ES2023 release हुआ

4. ES6 क्या है और इतना Important क्यों है?

ES6 को ES2015 भी कहा जाता है। यह JavaScript का एक बहुत बड़ा update था।

इसमें कई important features आए:

- `let` और `const`
- Arrow Functions
- Classes
- Promises
- Template Literals
- Destructuring
- Modules
- Default Parameters

उदाहरण:

Before ES6:

```
var name = "Morpal";

function greet() {
    return "Hello " + name;
}
```

After ES6:

```
const name = "Morpal";

const greet = () => {
    return `Hello ${name}`;
};
```

5. आज JavaScript कहाँ इस्तेमाल होती है?

Frontend Development
React, Angular, Vue

Backend Development
Node.js, Express, NestJS

Mobile Development
React Native

3D Web Development
Three.js, WebGL

6. Interview के लिए Important Questions

1. Who created JavaScript?
2. When was JavaScript created?
3. What was the original name of JavaScript?
4. What is ECMAScript?
5. What is the difference between Java and JavaScript?
6. What is ES6?
7. Why was JavaScript created?

Remember for Interview

JavaScript was created by Brendan Eich in 1995 at Netscape. Initially, it was named Mocha, then LiveScript, and finally JavaScript. ECMAScript is the standard specification on which JavaScript is based.

मेरी सलाह: JavaScript सीखते समय History के बाद इन topics को समझना शुरू करो: Execution Context → Call Stack → Hoisting → Scope → Closures → Event Loop → Promises → Async/Await. ये JavaScript के strong fundamentals हैं।

What is ECMAScript?

What is ECMAScript?

ECMAScript (ES) is a standard or specification that defines how JavaScript should work.

In simple words, ECMAScript provides the rules and guidelines that JavaScript follows.

1. Why was ECMAScript created?

- JavaScript was created by Brendan Eich in 1995.
- Different browsers started implementing JavaScript differently.
- To maintain consistency across browsers, a standard was needed.
- In 1997, ECMA International standardized JavaScript under the name ECMAScript.

2. JavaScript vs ECMAScript

JavaScript	ECMAScript
Programming language	Language specification
Used to build applications	Defines language rules
Implemented by engines	Defines features and behavior
Example: JS code	Example: ES6 standard

Remember: ECMAScript is the specification, and JavaScript is its implementation.

3. Important ECMAScript Versions

Version	Year	Important features
ES1	1997	First standard
ES3	1999	Regular expressions, try/catch
ES5	2009	Strict mode, JSON support
ES6 (ES2015)	2015	let, const, arrow functions, classes, promises
ES2017	2017	async/await
ES2020	2020	Optional chaining, nullish coalescing
ES2022	2022	Top-level await, private class fields
ES2024	2024	Object.groupBy, Promise.withResolvers

4. Example of ECMAScript features

These are JavaScript features introduced through ECMAScript standards.

ES5:

```
var name = "Morpal";

function greet() {
    return "Hello " + name;
}
```

ES6:

```
const name = "Morpal";

const greet = () => `Hello ${name}`;

console.log(greet());
```

5. Who maintains ECMAScript?

ECMA International

Organization responsible for standardizing ECMAScript.

Official Website

Interview Answer

“ECMAScript is a scripting language specification standardized by ECMA International. JavaScript follows the ECMAScript specification, which defines its syntax, features, and behavior. ES6, released

in 2015, introduced many important features like let, const, arrow functions, classes, and promises.”

Important concept

JavaScript ≠ ECMAScript

- ECMAScript = Rules / Standard
- JavaScript = Programming language following those rules
- V8 = JavaScript engine used by Chrome and Node.js

> ECMAScript kya haii

ECMAScript (ES) JavaScript ka ek official standard hai, jo define karta hai ki JavaScript language ka syntax, features aur behaviour kaise work karenge.

Simple language mein:

ECMAScript ek rulebook hai, jiske according JavaScript develop aur update hoti hai.

ECMAScript ki zarurat kyon padi?

1995 mein Brendan Eich ne JavaScript banayi. Alag-alag browsers JavaScript ko alag tarike se implement karne lage.

Is problem ko solve karne ke liye 1997 mein ECMA International ne JavaScript ko standardize kiya, jiska naam ECMAScript rakha gaya.

JavaScript vs ECMAScript

ECMAScript	JavaScript
Official specification	Programming language
Rules define karta hai	Rules follow karti hai
Features specify karta hai	Features implement karti hai

Example

ES6 (2015) mein ye features introduce hue:

```
// let and const
let age = 22;
const name = "Morpal";

// Arrow function
const greet = () => {
    console.log("Hello JavaScript");
};
```

```
// Template literals
console.log(`Hello ${name}`);
```

ECMAScript Versions

- ES1 – 1997
- ES3 – 1999
- ES5 – 2009
- ES6 (ES2015) – 2015
- ES2016 – 2016
- ES2020 – 2020
- ES2025 – 2025

Interview Definition

“ECMAScript is a standard specification that defines the syntax, features and behaviour of JavaScript. JavaScript is an implementation of the ECMAScript specification.”

Ek line mein yaad rakho:

ECMAScript = Rules

JavaScript = Language that follows those rules.

deeply explain create conten and exam interview our notes ke liye bhi

ECMAScript – Complete Notes (Beginner to Advanced)

(MCA Notes + Interview Preparation) (JavaScript Fundamentals)

ECMAScript ko deeply samajhna JavaScript developer ke liye bahut important hai. Agar tum React, Node.js, Next.js aur Full-Stack Development mein career banana chahte ho, toh ECMAScript ki understanding tumhare JavaScript fundamentals ko strong karegi.

Is topic ko hum 4 perspectives se samjhenge:

1. **Concept:** ECMAScript actually kya hai?
2. **History:** ECMAScript kyon bana aur kaise evolve hua?
3. **Technical:** ECMAScript, JavaScript, JavaScript Engine aur Browser ka relationship.
4. **Preparation:** Exam notes, interview questions aur practical examples.

---

1. What is ECMAScript?

Definition:

ECMAScript is a standardized scripting language specification that defines the syntax, semantics, data types, operators, objects, functions, and other core features of the JavaScript language.

Simple Hindi mein:

ECMAScript ek official specification (rulebook) hai jo define karti hai ki JavaScript language kaise behave karegi, uske rules kya honge aur usmein kaun-kaun se features hone chahiye.

For example:

- Variables kaise declare honge?
- Functions kaise work karenge?
- Objects kaise behave karenge?
- Arrays kaise work karengi?
- Promises ka behaviour kya hoga?
- Classes kaise define hongi?
- New language features kaise introduce honge?

In sabhi ke rules ECMAScript specification define karti hai.

Simple real-world analogy

ECMAScript

Rules and instructions

JavaScript

Language following the rules

Jaise cricket ke rules alag hote hain aur players un rules ke according game khelte hain, waise hi ECMAScript rules define karta hai aur JavaScript unhe implement karti hai.

2. History of ECMAScript

ECMAScript ko samajhne ke liye JavaScript ki history samajhna zaroori hai.

1995 – JavaScript was created

- Creator: Brendan Eich
- Company: Netscape
- Initially developed in approximately 10 days.
- Original name: Mocha
- Later name: LiveScript
- Final name: JavaScript

Problem: Different browsers JavaScript ko apne-apne ways mein implement karne lage. Isse browser compatibility issues hone lage.

Example:

Ek JavaScript program Netscape Navigator mein sahi work kar sakta tha, lekin Internet Explorer mein different behaviour de sakta tha.

Solution: Standardization

1996 mein JavaScript ko standardize karne ka process shuru hua. ECMA International ne language specification develop ki.

1997 mein first ECMAScript standard officially publish hua.

Important organizations

Publishes and maintains ECMAScript standards.

Technical committee responsible for developing ECMAScript language features.

3. What is TC39?

TC39 stands for Technical Committee 39.

It is a committee under Ecma International responsible for developing and maintaining ECMAScript.

Its main responsibilities:

1. Proposing new JavaScript features.
2. Evaluating feature proposals.
3. Discussing language improvements.
4. Maintaining the ECMAScript specification.
5. Standardizing approved features.

Example: Jab JavaScript mein optional chaining (`?.`) introduce karna tha, toh uske liye proposal process follow hua, review hua aur finally ECMAScript standard ka part bana.

4. JavaScript vs ECMAScript vs JavaScript Engine

Ye teen concepts interview mein bahut pooche jaate hain.

Concept	Meaning
ECMAScript	Language specification
JavaScript	Programming language
JavaScript Engine	Executes JavaScript code

JavaScript Engines examples:

- V8 – Chrome and Node.js
- SpiderMonkey – Firefox

- JavaScriptCore – Safari

How do they work together?

Developer writes JavaScript code

Example: const x = 10;

ECMAScript Specification

Defines language rules and expected behaviour

JavaScript Engine

Parses, compiles and executes JavaScript

Program Output

Expected result

Important clarification: ECMAScript specification directly code execute nahi karti. JavaScript engine specification ke rules ke according code implement aur execute karta hai.

---

5. ECMAScript Versions – Complete Evolution

ECMAScript time ke saath evolve hota raha hai. Har naye version mein language ke features improve hue.

Version	Year	Major changes
ES1	1997	First official specification
ES2	1998	Editorial updates
ES3	1999	Regular Expressions, try/catch
ES4	Abandoned	Proposed major changes, never standardized
ES5	2009	Strict Mode, JSON support, Array methods
ES5.1	2011	Minor specification update
ES6 / ES2015	2015	let, const, arrow functions, classes, promises
ES2016	2016	Exponentiation, includes()
ES2017	2017	async/await, Object.values()
ES2018	2018	Rest/Spread properties, Promise.finally()
ES2019	2019	flat(), flatMap(), optional catch binding
ES2020	2020	BigInt, Optional Chaining, Nullish Coalescing
ES2021	2021	replaceAll(), Logical assignment
ES2022	2022	Top-level await, private class fields

Version	Year	Major changes
ES2023	2023	findLast(), toSorted(), toReversed()
ES2024	2024	Object.groupBy(), Promise.withResolvers()
ES2025	2025	Iterator helpers, Set methods, RegExp.escape()

Important concepts:

- ES6 and ES2015 are the same version.
- ES6 was a major update in JavaScript history.
- After ES2015, ECMAScript started following an annual release cycle.
- ESNext is an informal term for upcoming or proposed JavaScript features.

Official version history: .

6. What exactly does ECMAScript define?

ECMAScript specification language ke multiple fundamental parts define karti hai.

A. Variables and Data Types

JavaScript mein variables declare karne ke different ways hain.

```
var name = "Morpal";

let age = 22;

const country = "India";
```

ECMAScript define karta hai ki `var`, `let` aur `const` ka behaviour kya hoga.

Data types:

```
// Primitive data types

let name = "Morpal";       // String
let age = 22;              // Number
let isStudent = true;      // Boolean
let x;                     // Undefined
let y = null;              // Null
let big = 123n;            // BigInt
let id = Symbol("id");     // Symbol

// Non-primitive

let user = { name: "Morpal" }; // Object
```

B. Functions

ECMAScript define karta hai ki functions kaise declare, call aur execute honge.

```
function add(a, b) {
    return a + b;
}

console.log(add(10, 20));
```

ES6 mein Arrow Functions introduce hue:

```
const add = (a, b) => a + b;

console.log(add(10, 20));
```

C. Objects and Arrays

Objects aur Arrays JavaScript ke important data structures hain.

```
const student = {
    name: "Morpal",
    course: "MCA",
    semester: 3
};

const skills = ["JavaScript", "React", "Node.js"];

console.log(student.name);
console.log(skills[0]);
```

D. Classes and Inheritance

ES6 mein class syntax introduce hua.

```
class Student {
    constructor(name) {
        this.name = name;
    }

    introduce() {
        console.log(`My name is ${this.name}`);
    }
}

const student = new Student("Morpal");

student.introduce();
```

E. Promises and Asynchronous Programming

Promises asynchronous operations ke result ko represent karte hain.

```
const myPromise = new Promise((resolve, reject) => {
    resolve("Data received successfully");
});

myPromise.then((result) => {
    console.log(result);
});
```

ES2017 mein Async/Await introduce hua.

```
async function getData() {
    const result = await myPromise;
    console.log(result);
}

getData();
```

F. Modules

ES Modules ki help se hum code ko multiple files mein divide kar sakte hain.

math.js

```
export const add = (a, b) => a + b;
```

app.js

```
import { add } from "./math.js";

console.log(add(10, 20));
```

Ye features React, Node.js aur modern JavaScript applications mein frequently use hote hain.

---

7. ECMAScript vs Web APIs – Important for Developers

Ye concept bahut important hai, especially jab tum JavaScript Event Loop aur asynchronous programming padhoge.

ECMAScript aur Browser Web APIs alag cheezein hain.

ECMAScript	Browser Web APIs
Variables	DOM
Functions	document
Objects	fetch()
Arrays	setTimeout()
Promises	localStorage

ECMAScript	Browser Web APIs
Classes	Browser events

Example:

```
// ECMAScript feature
const name = "Morpal";

const greet = () => {
    console.log(`Hello ${name}`);
};

// Browser Web API
document.getElementById("btn");
```

Yahan:

- `const` ECMAScript ka part hai.
- Arrow function ECMAScript ka part hai.
- `document` Browser DOM API ka part hai.

Important: `fetch()` aur `setTimeout()` ECMAScript ke core language features nahi hain. Browser mein ye Web APIs hain; Node.js mein similar APIs Node.js runtime provide karta hai.

8. ECMAScript and JavaScript Runtime

Ab ek important distinction samjho.

JavaScript Runtime = JavaScript Engine + Host Environment APIs and other runtime facilities.

Examples:

Browser Runtime

V8 (Chrome), DOM, Web APIs, event loop integration

Node.js Runtime

V8, Node APIs, libuv and event loop integration

Example:

```
console.log("Start");

setTimeout(() => {
    console.log("Timer");
}, 1000);

console.log("End");
```

Output:

```
Start
End
Timer
```

Yahan `setTimeout()` host environment ki API hai aur callback scheduling runtime ke event-loop mechanisms se coordinate hoti hai.

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9. ECMAScript Standards ka Proposal Process (TC39)

New JavaScript feature directly language mein add nahi hota. Uske liye proposal process hota hai.

TC39 proposal ke generally 5 stages hain:

1. **Stage 0 – Strawperson**

   Initial idea or suggestion.

2. **Stage 1 – Proposal**

   Problem aur solution ko formally explore kiya jata hai.

3. **Stage 2 – Draft**

   Feature ka detailed specification draft prepare hota hai.

4. **Stage 3 – Candidate**

   Feature implementation aur testing ke liye mature hota hai.

5. **Stage 4 – Finished**

   Requirements complete hone par feature standard mein include hone ke liye ready hota hai.

Note: Stage 4 feature ko final specification mein include karne ke liye ready karta hai; official edition mein publication alag step hai.

Official proposals: .

---

10. ECMAScript Important Features with Examples

Ye table tumhare revision ke liye important hai.

Feature	Version	Example
let / const	ES6	`const age = 22`
Arrow Function	ES6	`() => {}`

Feature	Version	Example
Template Literals	ES6	`` `Hello ${name}` ``
Destructuring	ES6	`const {name} = user`
Spread Operator	ES6	`[...arr]`
Rest Parameters	ES6	`(...args) => {}`
Classes	ES6	`class Student {}`
Promises	ES6	`new Promise()`
Async/Await	ES2017	`async function`
Optional Chaining	ES2020	`user?.name`
Nullish Coalescing	ES2020	`value ?? 10`
BigInt	ES2020	`123n`
Private Fields	ES2022	`#password`
Object.groupBy	ES2024	`Object.groupBy()`

Part 2: Exam Preparation Notes

Ye section MCA semester exams aur written examinations ke liye hai.

11. ECMAScript – 5 Marks Answer

Question: What is ECMAScript? Explain its history and importance.

Answer:

ECMAScript is a standardized scripting language specification developed and maintained under ECMA International. It defines the rules, syntax, semantics and features of JavaScript.

ECMAScript was introduced in 1997 to standardize JavaScript and ensure consistent behaviour across different browsers and platforms.

It is maintained by Technical Committee 39 (TC39).

History:

1. JavaScript was created by Brendan Eich in 1995.
2. Initially, it was named Mocha.
3. Later, it was renamed LiveScript.
4. Finally, it became JavaScript.
5. In 1997, ECMAScript 1 was officially standardized.

Importance:

- Provides standardization.
- Ensures consistent language behaviour.
- Defines syntax and data types.
- Introduces new programming features.
- Improves compatibility between JavaScript implementations.

Conclusion: ECMAScript provides the foundation and standard rules for modern JavaScript development.

12. ECMAScript – 10 Marks Answer

Question: Explain ECMAScript in detail. Discuss its versions and features.

Answer:

ECMAScript is an international scripting language specification that defines the behaviour and features of JavaScript. It is standardized by ECMA International and developed by TC39.

History

JavaScript was created by Brendan Eich in 1995 at Netscape. Initially, it was called Mocha, then LiveScript and finally JavaScript. In 1997, the first ECMAScript standard was released.

Major Versions

- ES1 (1997): First standard.
- ES3 (1999): Regular expressions and exception handling.
- ES5 (2009): Strict mode and JSON support.
- ES6 (2015): let, const, arrow functions, classes and promises.
- ES2017: Async/Await.
- ES2020: Optional chaining and BigInt.
- ES2022: Private class fields and top-level await.

Features

1. Dynamic typing.
2. Object-oriented programming support.
3. Functions and closures.
4. Promises and asynchronous programming.
5. Classes and inheritance.
6. Modules.
7. Modern operators and syntax.

Importance

ECMAScript provides standardization, consistency, compatibility and continuous improvements to JavaScript.

Conclusion: ECMAScript is the foundation of modern JavaScript and plays an important role in frontend and backend development.

Part 3: Interview Preparation

13. Top 15 Interview Questions

Interview question bank (/ viewed)

14. Test Your Knowledge – MCQ

ECMAScript Quiz (7 Questions)

Part 4: Quick Revision Notes

ECMAScript – One Page Revision

Topic	Remember
Creator of JS	Brendan Eich
Created in	1995
First ECMAScript	1997
Standard body	ECMA International
Technical committee	TC39
ES6	ES2015
ES6 major features	let, const, arrow functions, classes
JavaScript engine	V8, SpiderMonkey, JavaScriptCore
ESNext	Upcoming/proposed features

Three lines to remember:

1. ECMAScript is a specification.
2. JavaScript is a language that implements ECMAScript.
3. JavaScript engines execute JavaScript code.

[Copy revision notes]

Recommended learning sequence

ECMAScript samajhne ke baad ye order follow karna. Isse JavaScript ke concepts ek doosre se connect honge:

1. Variables: `var`, `let`, `const`
2. Data Types and Type Coercion
3. Operators and Control Flow
4. Functions and Arrow Functions
5. Scope and Lexical Environment
6. Hoisting and Temporal Dead Zone
7. Closures
8. `this` keyword
9. Objects and Prototypes
10. Classes and Inheritance
11. Execution Context and Call Stack
12. Promises and Async/Await
13. Event Loop
14. Modules
15. Advanced JavaScript

My recommendation: Har topic ke liye three things prepare karo: Definition, Internal Working aur Code Example. Is approach se MCA exams ke written answers aur software developer interviews dono ki preparation hogi.